Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Canon Morrow	canonm@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Website that takes in chemical formulas and gives the option to find the atomic number of the whole molecule, atomic mass of a part of it, or the charge of the molecule. It will also have a periodic table page, where the user can find basic information about elements from an interactive list of elements and common polyatomics.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

webqc.org - molar mass calculator thisperiodictable.com -visual periodic table with basic info

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My site will act as a one-stop website that will be easier to use for students by combining the resources and the calculator into one basic site.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Take elemental input, process the pathway or calculation, and present the atomic mass or final basic information.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Elemental formula for a molecule. Or an inputted path of clicks to guide toward info

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will take the formula, split it into individual elemental atoms, Compare it to a database of all elemental molar masses, multiply the individual element by its given multiplier, (ie O2= molO*2), adds up all the individual molar masses, and prints the final mass. Or, it will take in the inputted pathway and use a basic page chain to get to the right information

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The final molar mass of a molecule or basic info about an element

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

“Molar Mass”, “Element”, “Molecule”, "Multiplier", “Moles Present”

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Needed: Element name, element weight, elemental charge *Ignored: Atomic Number, Matter_State, Electronegativity*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The actual calculation of the molar mass

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will check it line by line manually

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Build a tool for another class

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool to help students enter chemical formulas and calculate total molar mass, the mass contributed by specific elements, and the charge of known molecules or ions. The website will also include an interactive periodic table and a reference list of common polyatomic ions. What are 3 common problems students face when working with chemical formulas and calculating molar mass, and what are 3 simple features a basic chemistry tool should have to help solve these problems?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

1. Three common problems students face

1. Calculating molar mass incorrectly: Students may forget to multiply an element's atomic mass by its subscript. For example, in H_2O, hydrogen's atomic mass must be multiplied by 2.

2. Understanding chemical formulas and element contributions: Students may struggle with formulas containing parentheses, such as $Ca(OH)_2$, or figuring out how much each element contributes to the total molar mass.

3. Finding reliable information about elements and ion charges: Students may have trouble finding atomic numbers, atomic masses, and the correct charges of common polyatomic ions.

2. Three simple features the tool should have

1. Chemical formula calculator: Users enter a chemical formula and receive its total molar

mass, with a breakdown showing each element's contribution.

2. Interactive periodic table: Users select an element to view its name, symbol, atomic number, and atomic mass.

3. Polyatomic ion reference table: Users browse common polyatomic ions to find their formulas and charges, such as sulfate (SO_4^{2-}), nitrate (NO_3^-), and ammonium (NH_4^+).

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

would feature 2 of the interactive periodic table be too complex for an intro to comp sci python class becuase i would need to make both a UI that is interactive and aesthetic?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I have decided to add reference sheets for items such as polyatomics and diatomics on the calculator, as well as put the information page as a chart rather than a list

Part 6: Resource Planning

1. What resources will you use if you get stuck?

The LLM, The CISC discord, Python community boards online

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Run it through Thonny's debug tool and follow the code step by step from a found safe starting point to find where the bug is and fix it.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Website that takes in chemical formulas and gives the option to find the atomic number of the whole molecule, atomic mass of a part of it, or the charge of the molecule. It will also have a periodic table page, where the user can find basic information about elements from an interactive list of elements and common polyatomics.

This is a useful chemistry project idea for a CS1 final project! Based on your description, I'd recommend keeping the main features focused on chemical formula calculations and an interactive reference table so the application is realistic to build in Drafter within three weeks.

Suggested project plan

Project description (polished):

A chemistry website that allows users to enter chemical formulas and calculate the total molar mass, the mass contributed by specific elements, and the overall charge of a molecule or ion. The website will also include an interactive periodic table where users can explore basic information about elements, along with a reference list of common polyatomic ions.

Recommended features

Feature	What it does	Difficulty
Chemical formula input	Accepts formulas such as `H2O` and `Ca(OH)2`	Medium
Molar mass calculator	Adds the atomic masses of all atoms in a formula	Medium

Element mass calculator	Calculates how much a particular element contributes to the total molar mass	Medium
Charge information	Displays the charge of a known ion or formula from a reference list	Easy–Medium
Interactive periodic table	Displays element names, symbols, atomic numbers, and atomic masses	Easy–Medium
Polyatomic ion reference	Lists common ions, their formulas, and their charges	Easy

One important chemistry clarification

- Atomic number: Belongs to an individual element, not a whole molecule. For a molecule, you could instead calculate the sum of the atomic numbers of all its atoms, clearly labeling it as a custom calculation.
- Molar mass: The mass of one mole of a compound, typically expressed in g/mol. This is probably the most useful calculation for your website.
- Element contribution: For example, in H_2O, hydrogen contributes approximately 2.016 g/mol and oxygen contributes approximately 15.999 g/mol.
- Charge: A formula alone does not always tell you the charge unless the formula represents a known ion or includes a charge notation. Your site should use a known-ion reference list or ask the user for the charge when needed.

My recommendation: Build the periodic table, polyatomic ion reference, and molar mass calculator first. Add formula parsing for parentheses and more complicated formulas only after the basic calculations work. Avoid trying to build a complete chemistry engine or automatically determine the charge of any arbitrary compound.

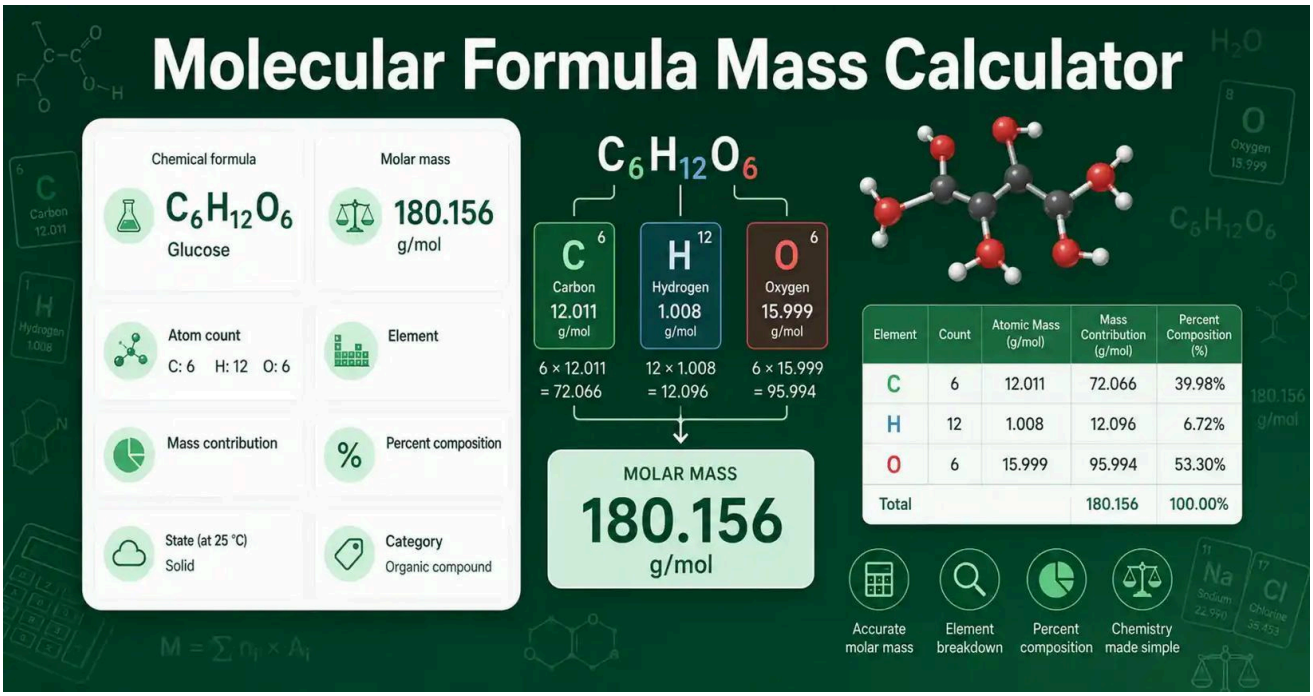
This approach fits Drafter well because you can store element and ion information in Python dictionaries and lists, perform calculations locally, and display results without needing a backend or database.

are there any websites or tools that currently adress this

Yes! There are several existing websites that address parts of your chemistry website idea. These are useful for Part 1, Question 2: Prior Solutions in your project worksheet.

The main opportunity for your project is to combine formula-based calculations, an interactive periodic table, and a common polyatomic ion reference into one student-friendly website.

Existing websites to research

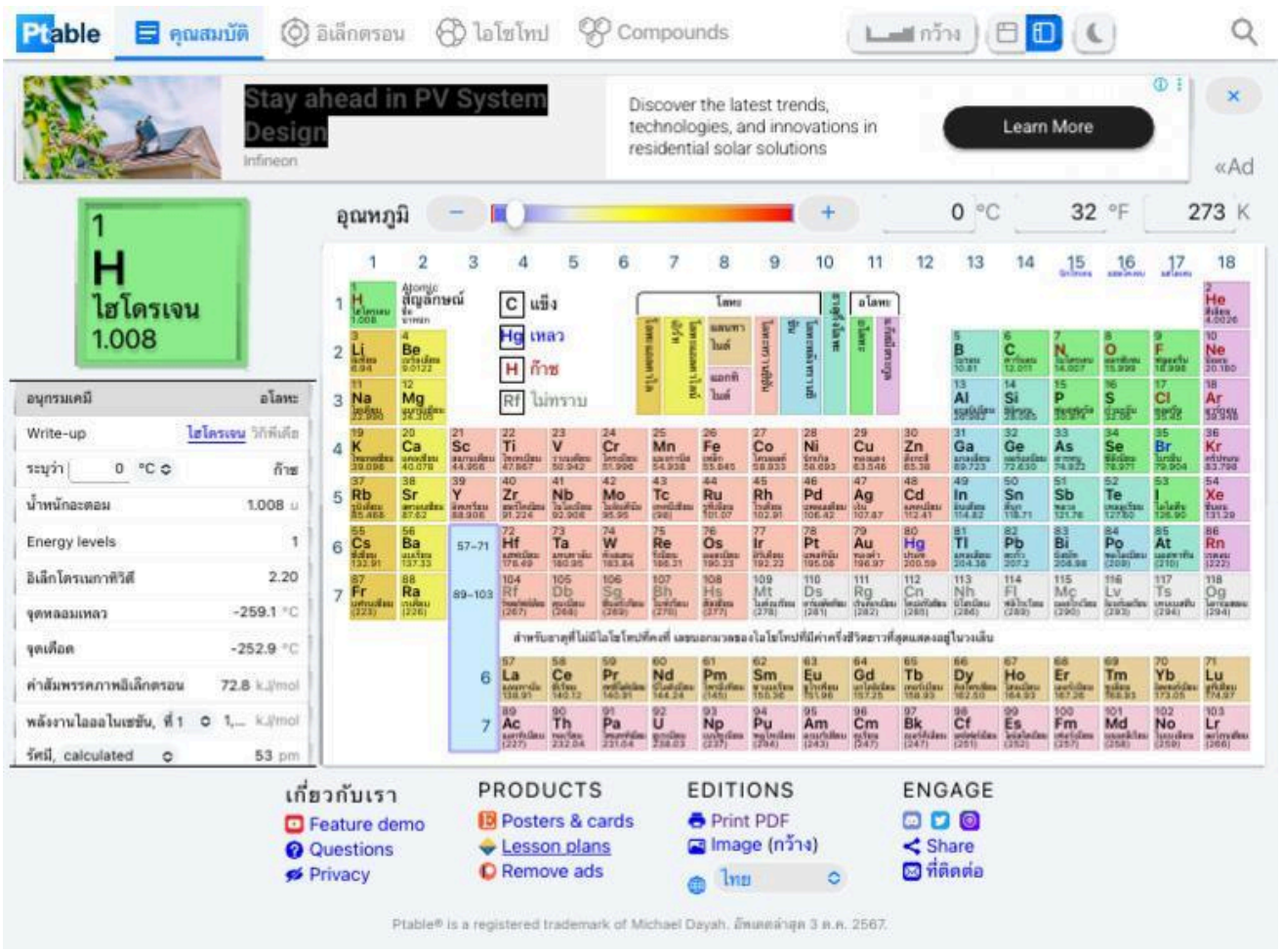


1. WebQC

Closest match for calculations

Website: webqc.org

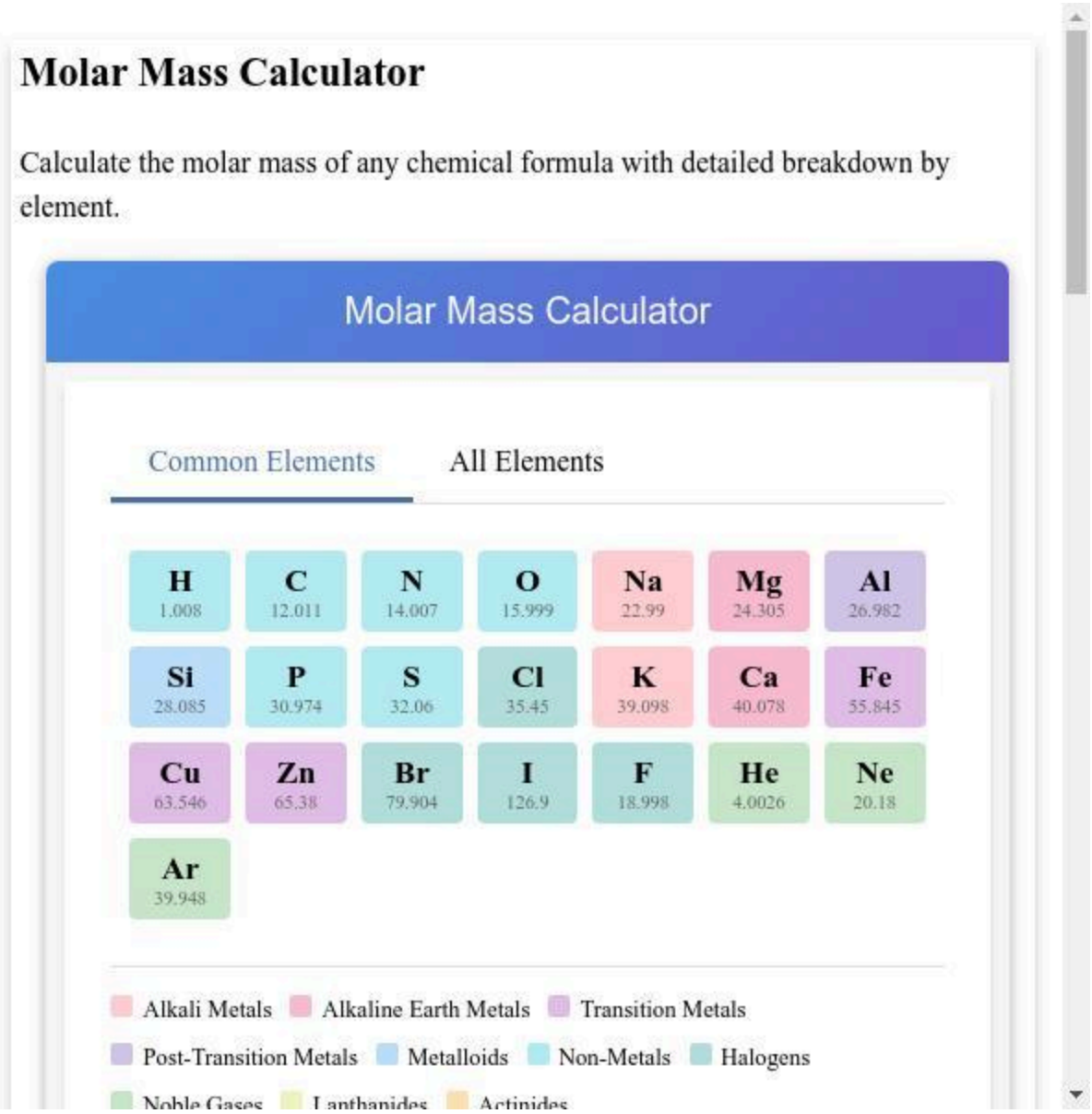
Accepts chemical formulas and calculates molar mass, elemental composition, and each element's mass contribution. It also supports formulas with parentheses.【turn440847search0】



2. Ptable

Website: ptable.com

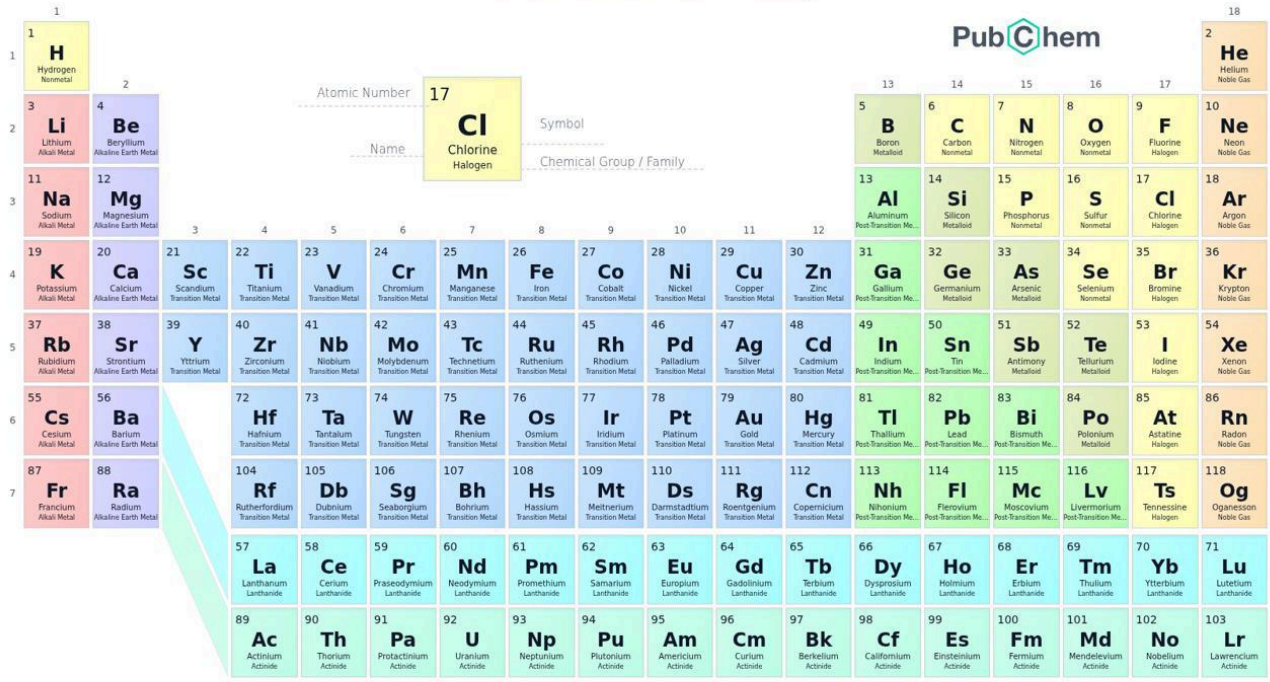
An interactive periodic table for exploring elements and their properties. This is a good example to study for your periodic table page.



3. This Periodic Table

Website: thisperiodictable.com

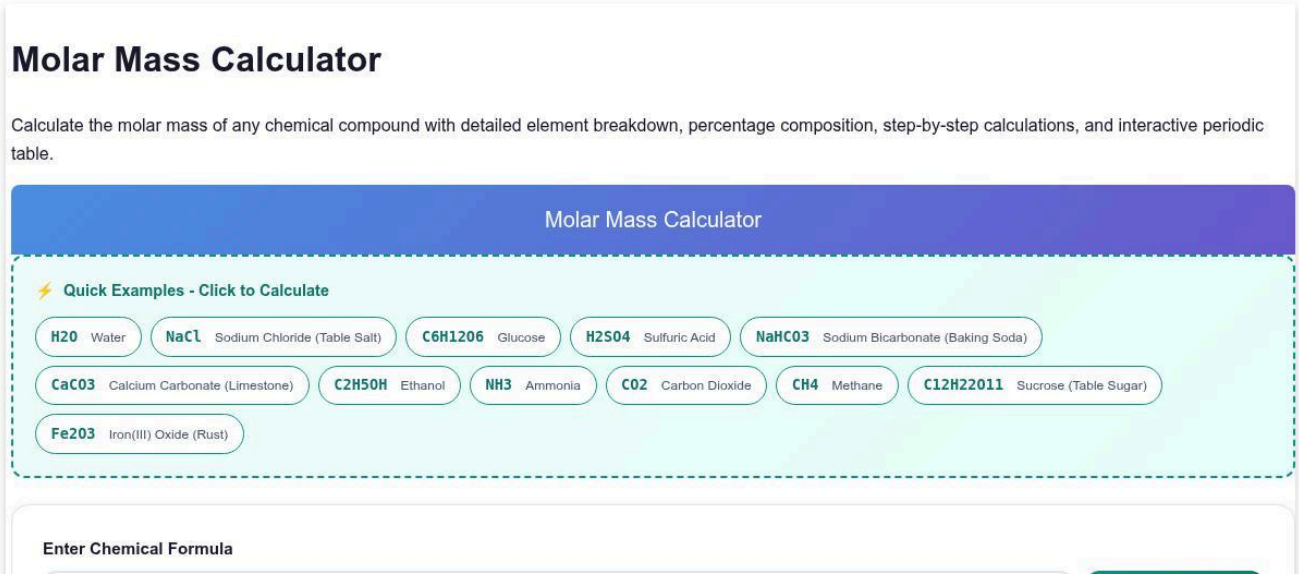
Combines a visual periodic table with a molar mass calculator, including the ability to hide element names or symbols as a learning aid.【turn440847search3】



4. PubChem

Website: pubchem.ncbi.nlm.nih.gov

Provides chemical information, compound properties, and molecular data. It is more of a chemical information resource than a simple student calculator.



5. PeriodicElementTable.com

Website: periodicelementtable.com/tools

Offers chemistry calculators for molar mass, percent composition, and other formula-related calculations, along with element information.【turn440847search7】

What does this mean for your project?

Your idea is not completely new, but that is normal for a good CS1 project. You can differentiate it by making the features especially easy for introductory chemistry students to use.

Feature	Already available elsewhere?
Molar mass from a formula	Yes
Element-by-element mass contribution	Yes
Interactive periodic table	Yes

Polyatomic ion reference list	Common chemistry reference material
Charge lookup integrated with formula tools	Some chemistry tools offer related functionality
All your chosen features in one simple, student-friendly interface	A reasonable direction for your project

Suggested answer for your worksheet

Several existing websites provide chemistry tools similar to my project idea. WebQC calculates molar mass and the mass contribution of each element in a chemical formula. Ptable provides an interactive periodic table where users can explore elements and their properties. This Periodic Table combines a periodic table with a molar mass calculator, and PubChem provides detailed information about chemical compounds. My project would aim to combine a formula calculator, an interactive periodic table, and a reference list of common polyatomic ions in one simple website designed for students studying introductory chemistry.

My advice: Explore WebQC and This Periodic Table first. Try entering `H2O`, `NaCl`, and `Ca(OH)2` to see how they display results. Then decide what your version could make simpler or more useful for students.

This might also interest you.

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"I want to build a simple web tool to help enter chemical formulas and calculate the total molar mass, the mass contributed by specific elements, and the overall charge of a molecule or ion. The website will also include an interactive periodic table where users can explore basic information about elements, along with a reference list of common polyatomic ions.. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?

Your prompt has a small mix-up: the questions about tracking time are from the example in your assignment. Since your project is about chemistry, your questions should focus on

chemical formulas and chemistry tools instead.

Here is a corrected version you can use for Part 5, Question 1 of your worksheet.

I want to build a simple web tool to help students enter chemical formulas and calculate total molar mass, the mass contributed by specific elements, and the charge of known molecules or ions. The website will also include an interactive periodic table and a reference list of common polyatomic ions. What are 3 common problems students face when working with chemical formulas and calculating molar mass, and what are 3 simple features a basic chemistry tool should have to help solve these problems?

This version is focused on your actual project and should help you brainstorm realistic features for your three-week CS1 assignment.

This might also interest you.

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I want to build a simple web tool to help students enter chemical formulas and calculate total molar mass, the mass contributed by specific elements, and the charge of known molecules or ions. The website will also include an interactive periodic table and a reference list of common polyatomic ions. What are 3 common problems students face when working with chemical formulas and calculating molar mass, and what are 3 simple features a basic chemistry tool should have to help solve these problems?

1. Three common problems students face

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2. Three simple features the tool should have

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number, and atomic mass.

3. Polyatomic ion reference table: Users browse common polyatomic ions to find their formulas and charges, such as sulfate (SO_4^{2-}), nitrate (NO_3^-), and ammonium (NH_4^+).

3. Summary for your worksheet

Three common problems students face when working with chemical formulas are calculating molar mass incorrectly, understanding how subscripts and parentheses affect calculations, and finding accurate information about elements and ion charges. Three useful features for my website would be a chemical formula calculator that displays total molar mass and individual element contributions, an interactive periodic table that shows basic element information, and a reference table of common polyatomic ions with their formulas and charges. These features would help students check their work and better understand introductory chemistry concepts.

Scope tip: For a three-week CS1 project, start with formulas that contain element symbols and subscripts, then add parentheses if time permits. For ion charges, use a reference table of known ions rather than trying to automatically determine the charge of every possible compound.

This might also interest you.

Ad
Ad options
Learn more

would feature 2 of the interactive periodic table be too complex for an intro to comp sci python class becuase i would need to make both a UI that is interactive and aesthetic?

I think an interactive periodic table is manageable for an introductory computer science Python class, as long as you simplify the design. The challenging part isn't necessarily the chemistry data or Python logic; it's trying to make the table look polished, interactive, and easy to use all at once.

Since you're building with Drafter, I'd recommend making the periodic table a simpler feature rather than trying to recreate a professional website like Ptable.

How difficult would it be?

1. Storing element information

Easy

Store element names, symbols, atomic numbers, and atomic masses in Python dictionaries or lists. You don't need a database.

2. Making it interactive

Moderate

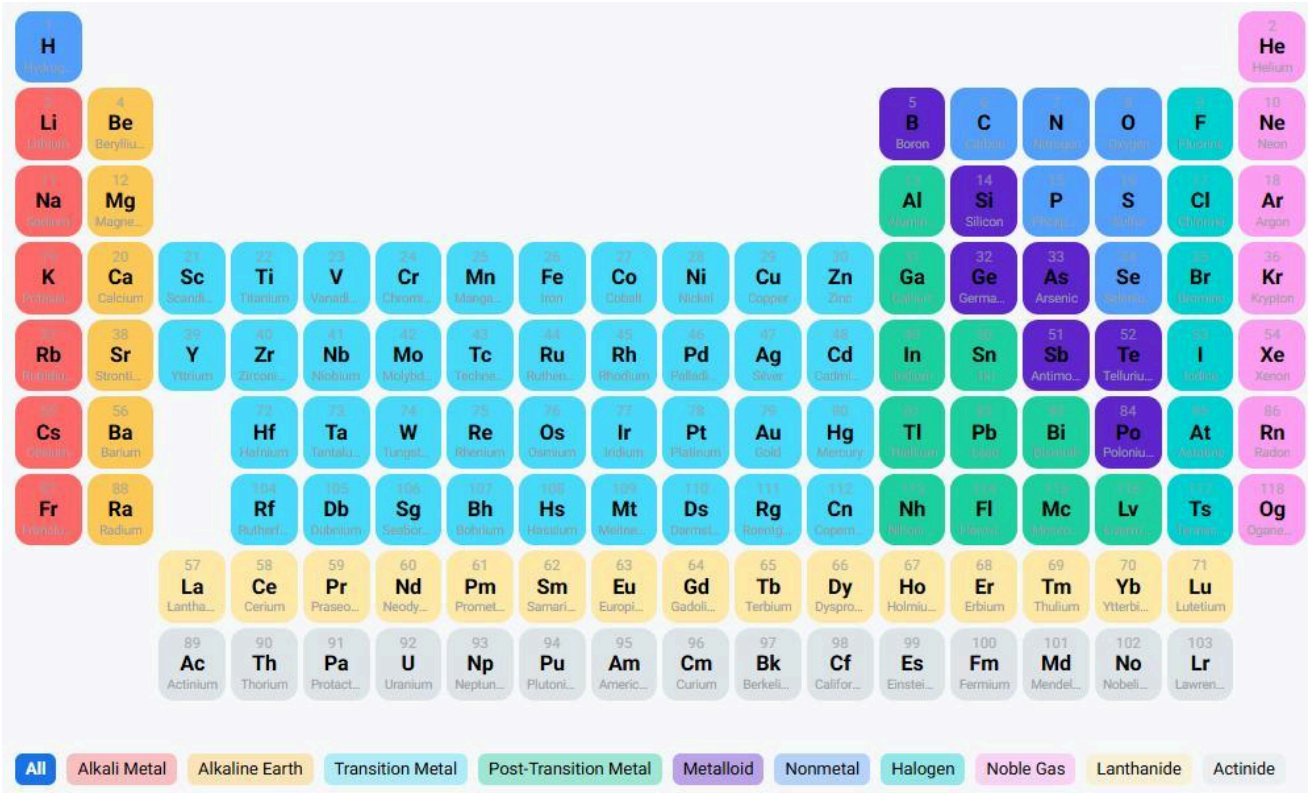
You could use buttons for elements. When a user clicks one, the page displays its information. This is much simpler than building custom animations or hover effects.

3. Making it aesthetic

Moderate

Consistent colors, spacing, and readable text can make the page look good without complex styling. Recreating all 118 elements in the exact periodic table layout will take more work.

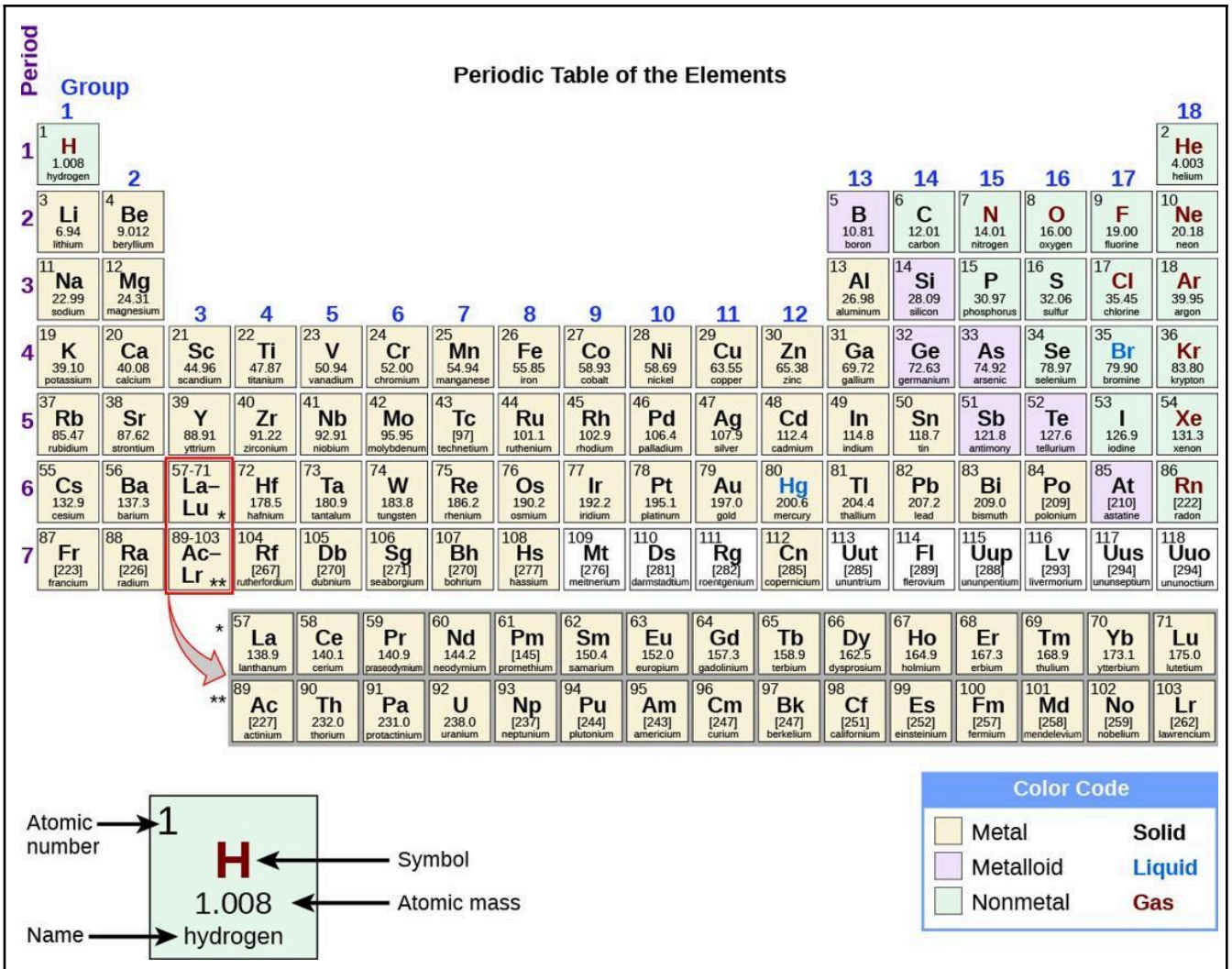
Three possible designs



Option A: Element grid — easiest

Show elements as colored buttons in a regular grid. Clicking a button reveals the element's name, atomic number, symbol, and atomic mass.

My recommendation



Option B: Traditional periodic table — moderate

Arrange elements in their familiar rows and columns, with different colors for element categories. Clicking an element displays its information.

This looks more authentic, but accurately positioning all 118 elements takes additional planning.



Option C: Searchable element directory — easiest

Let users search by name or symbol, or select an element from a list. Display its properties in a clean information card.

This is the safest choice if your formula calculator is the main priority.

What I would build for your project

I'd choose Option A, or Option B if the traditional periodic table layout is important to you.

Keep the interaction simple:

1. Display element buttons with their symbols and atomic numbers.
2. Give element categories a small, consistent color palette.
3. When a user clicks an element, display a detail card with its name, atomic mass, and atomic number.
4. Add a simple legend explaining the colors.

You do not need animations, custom hover effects, or a complicated responsive layout to make it look good.

Keep your three-week timeline in mind

Priority	Feature	Recommendation
1	Molar mass calculator	Core feature; build first
2	Polyatomic ion reference	Simple, useful lookup table
3	Interactive periodic table	Start with a small grid, then expand
4	Advanced styling and animations	Skip unless everything else works

One important Drafter consideration: its standard components may not support every kind of custom layout or styling you see on chemistry websites. A traditional periodic table with precisely positioned cells may require more layout work than a simple grid of buttons.

My verdict: Don't abandon the periodic table. Just avoid making the design itself a second major programming project. A clean grid of clickable elements is a reasonable CS1 feature, and it can still look polished.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.